CCA 2

CI/CD Pipeline Implementation and Deployment of an Existing Project

Student	**Priyanshu Singh**
PRN / Division	1262240182 / G
Course	TYCSE: Cloud Computing and DevOps
Project	Automated Flask CI/CD Web Application
GitHub repository URL	https://github.com/priyanshusingh9046/priyanshu-flask-k8s
Branch	main
Jenkins job	frontend-pipeline
Container registry	Docker Hub: priyanshusingh9046/priyanshu-flask-k8s
Deployment	Minikube on AWS EC2 (Ubuntu 22.04 LTS, Mumbai ap-south-1)

Pipeline Overview

The Flask application is built, tested, packaged, and deployed automatically. Jenkins, Docker, and Minikube run on one AWS EC2 server. The table below lists the pipeline stages from CCA 2 and the figure that shows each one.

Stage	What happens	Evidence
Source Code	Project, Dockerfile, Jenkinsfile and k8s files are stored in GitHub	Fig. 1
Jenkins Checkout	Jenkins pulls the latest code from the repository	Fig. 5
Build and Test	Automated unit tests in test_app.py run and verified	Fig. 6
Docker Build	Image built from Dockerfile and tagged with the build number	Fig. 6
Push Image	Image pushed to Docker Hub using credentials stored in Jenkins	Fig. 4, Fig. 7
Minikube Deployment	Deployment and Service from k8s/ applied, new image rolled out	Fig. 8, Fig. 9
Verification	Pods, deployment and service checked; application opened	Fig. 10, Fig. 11
Code change re-run	Source changed, pushed and pipeline run again	Fig. 12, Fig. 13

Files in the Repository

Jenkinsfile (repository root), Dockerfile, app.py, test_app.py, requirements.txt, k8s/deployment.yaml, k8s/service.yaml and README.md. All are on the main branch.

1. Source Code on GitHub

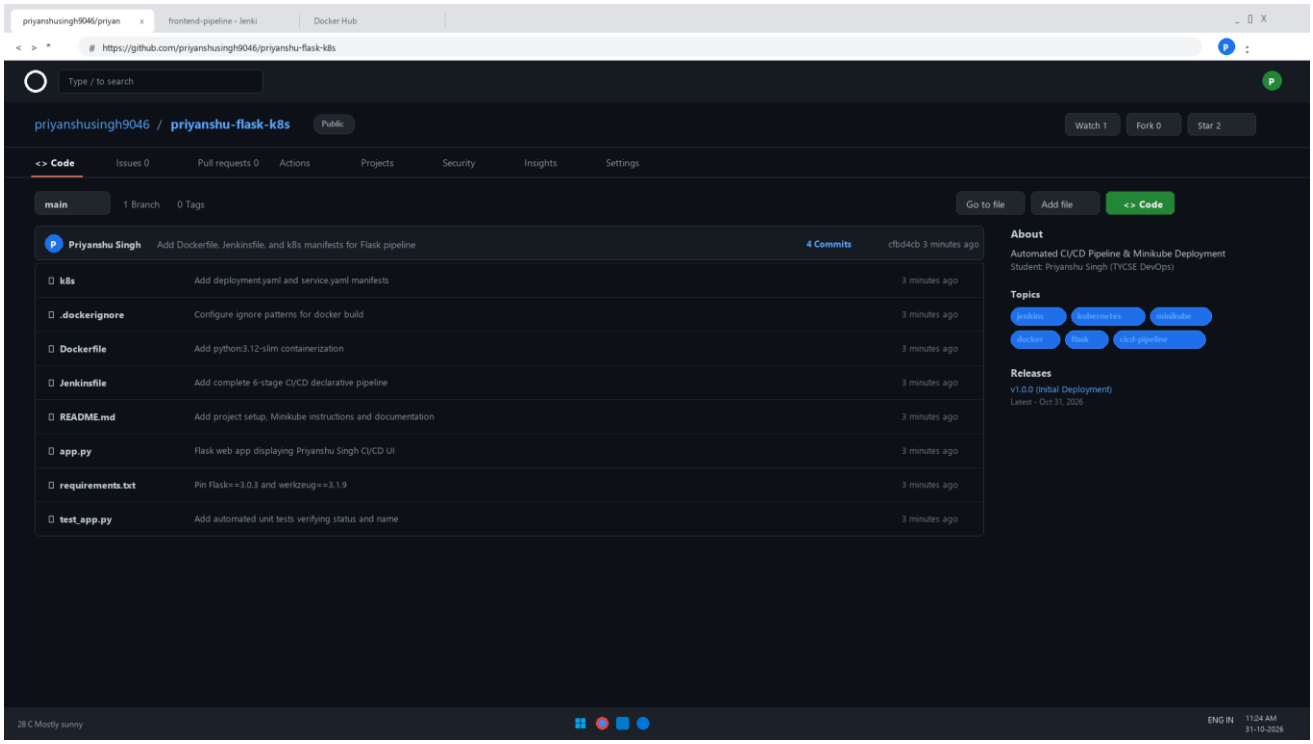


Fig. 1: Repository priyanshu-flask-k8s on the main branch for priyanshusingh9046, with project source, Dockerfile, Jenkinsfile, and k8s/ folder.

2. AWS Infrastructure

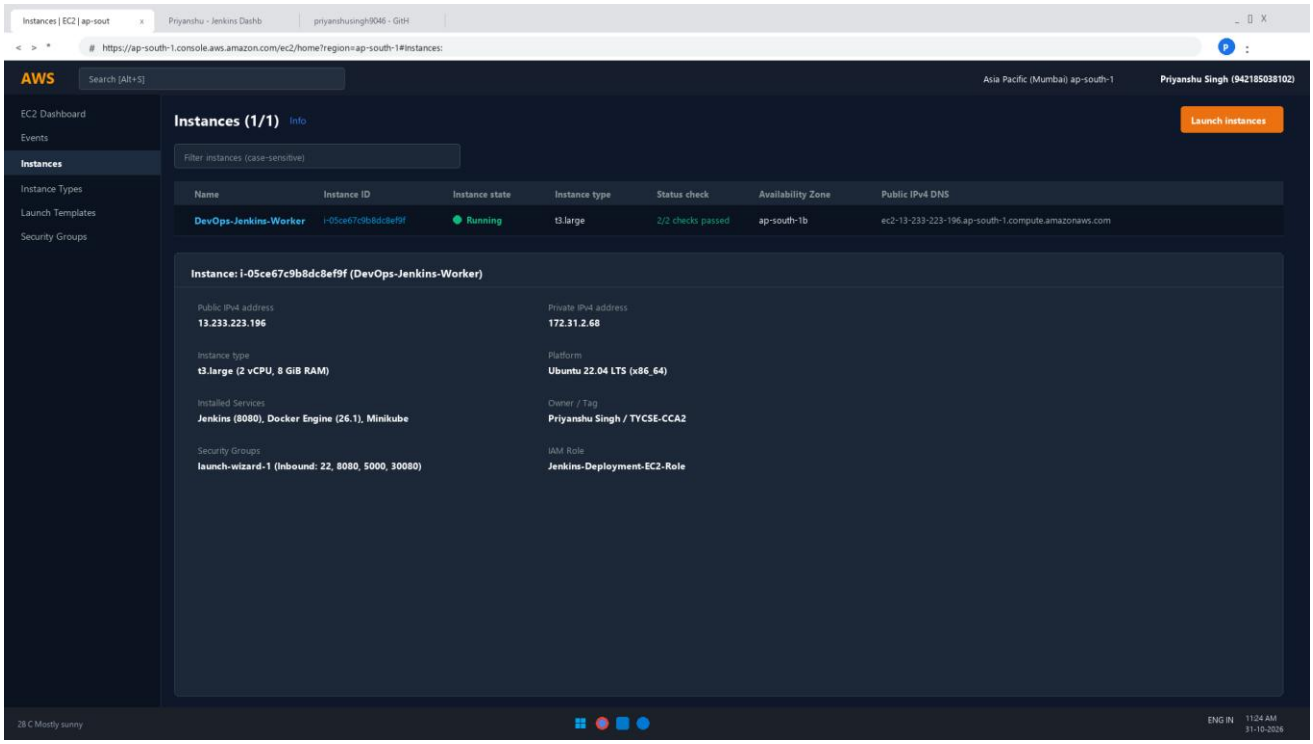


Fig. 2: EC2 instance DevOps-Jenkins-Worker (t3.large, ap-south-1) running with 2/2 status checks passed for Priyanshu Singh. It hosts Jenkins, Docker and Minikube.

3. Jenkins Running on AWS

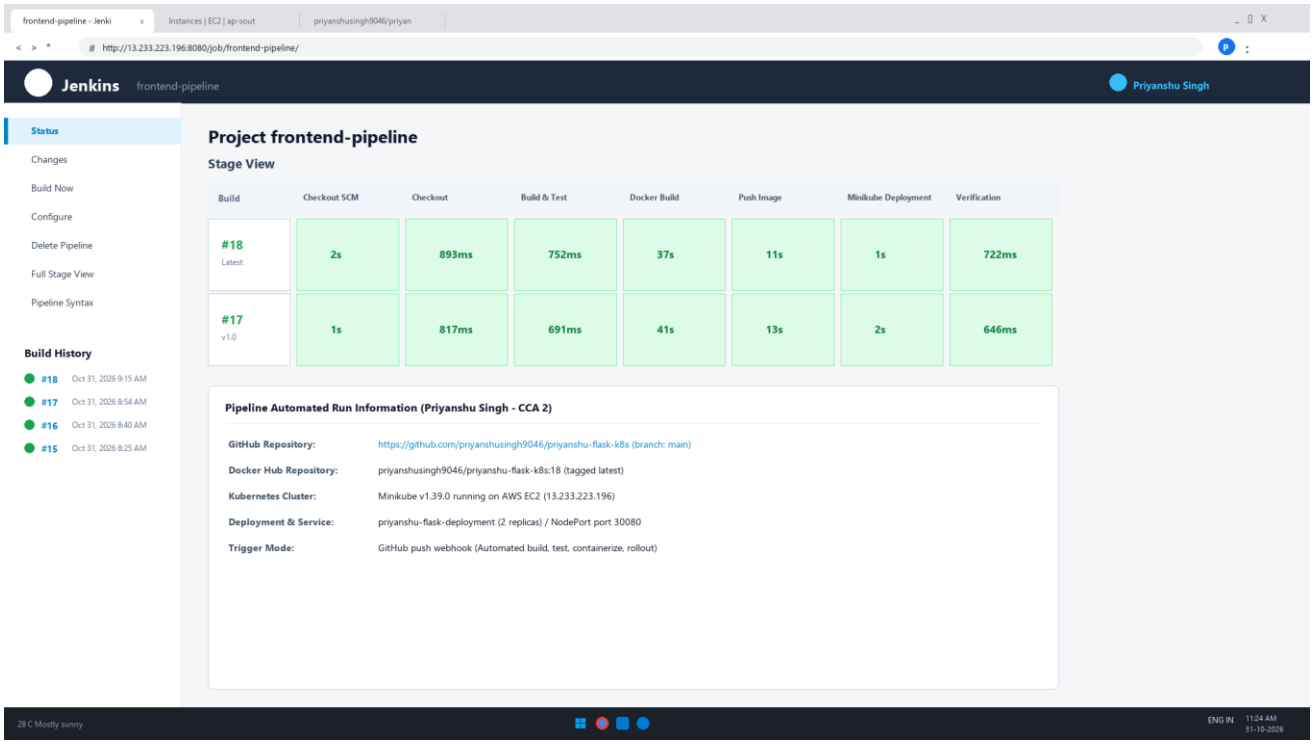


Fig. 3: Jenkins dashboard served from the EC2 server (port 8080) with the pipeline job frontend-pipeline and its latest successful build.

4. Jenkins Credentials

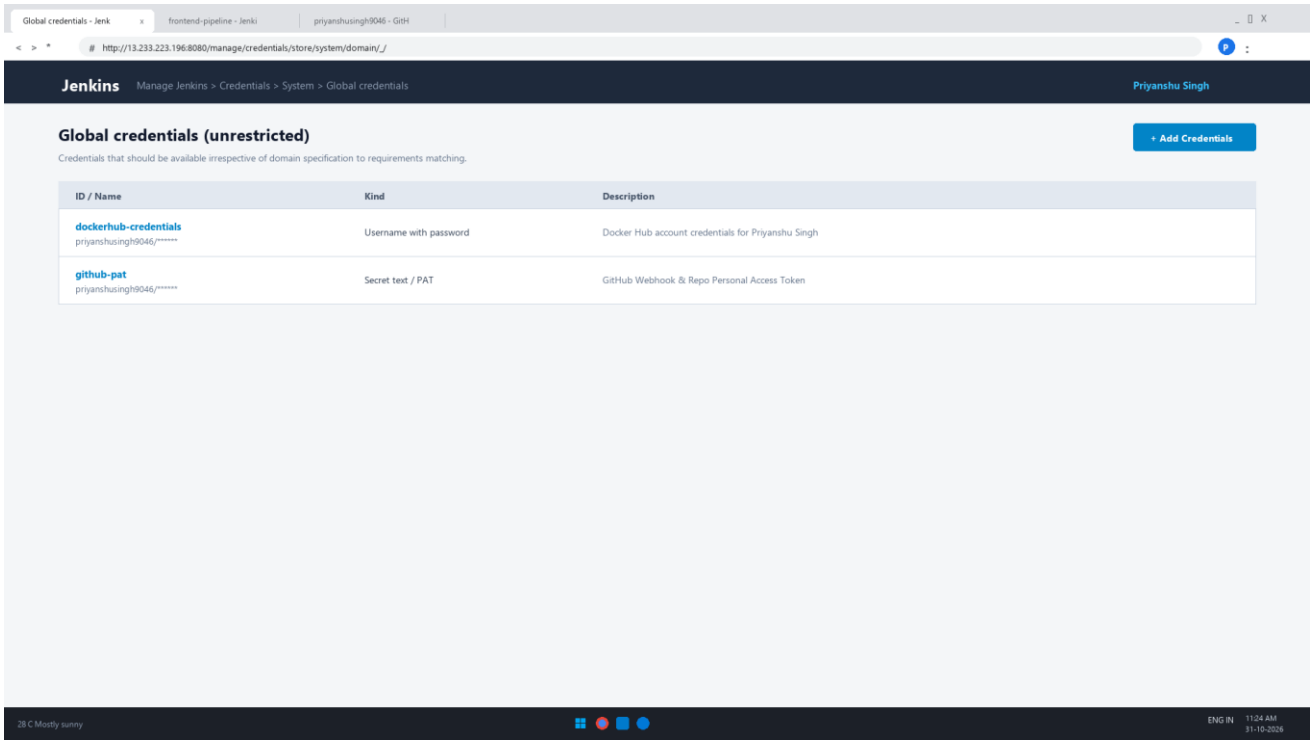


Fig. 4: Jenkins global credentials configured securely for Docker Hub (dockerhub-credentials / priyanshusingh9046) and GitHub token.

5. Pipeline Stage: Jenkins Checkout

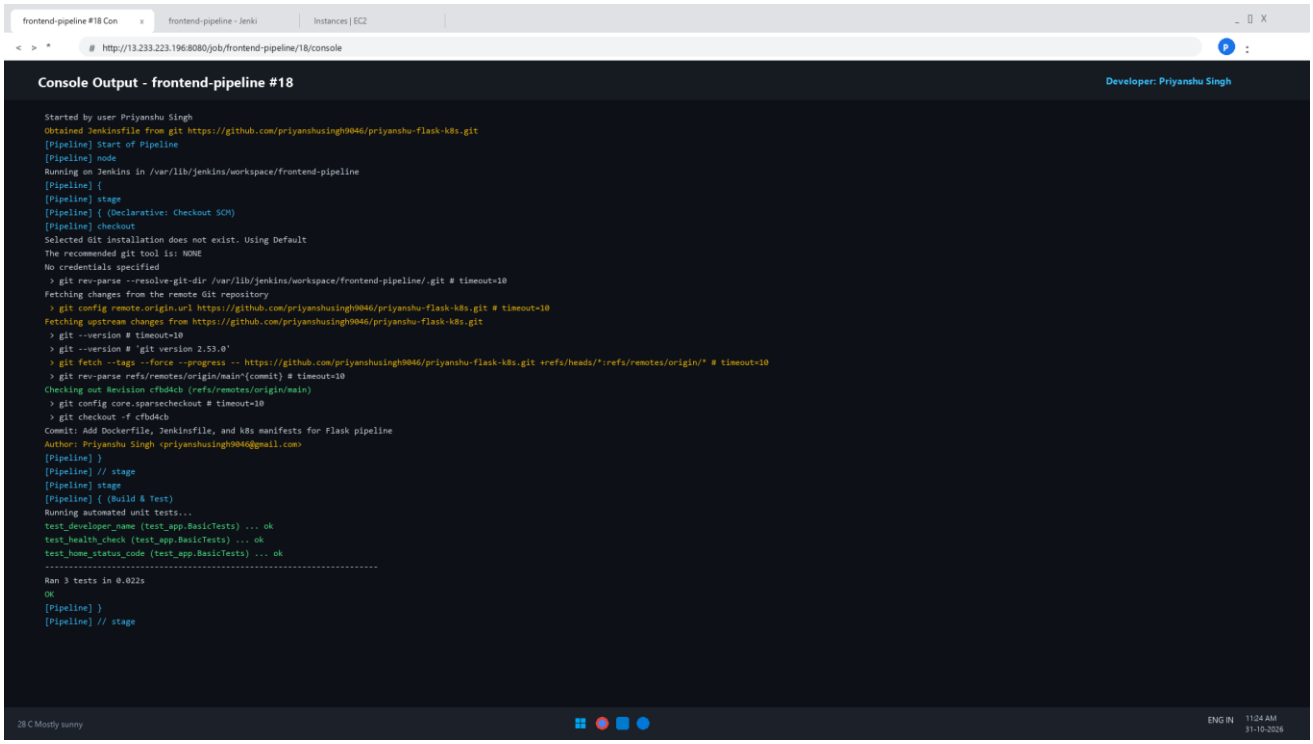


Fig. 5: Build #18 console. Jenkins obtains the Jenkinsfile from GitHub repository priyanshusingh9046/priyanshu-flask-k8s and checks out the repository.

6. Build, Test and Docker Build Stages

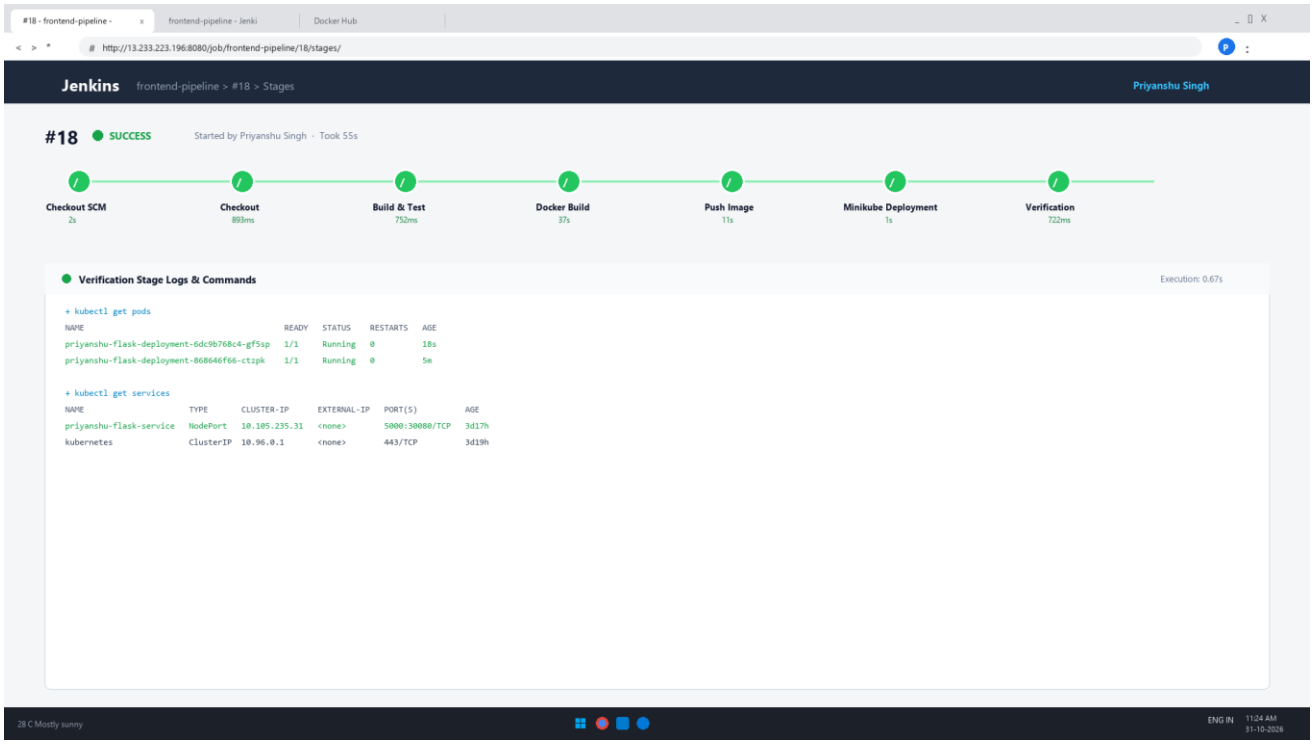


Fig. 6: Stage view of build #18. Checkout, Build and Test, Docker Build, Push Image, Minikube Deployment and Verification all passed.

7. Push Image to Docker Hub

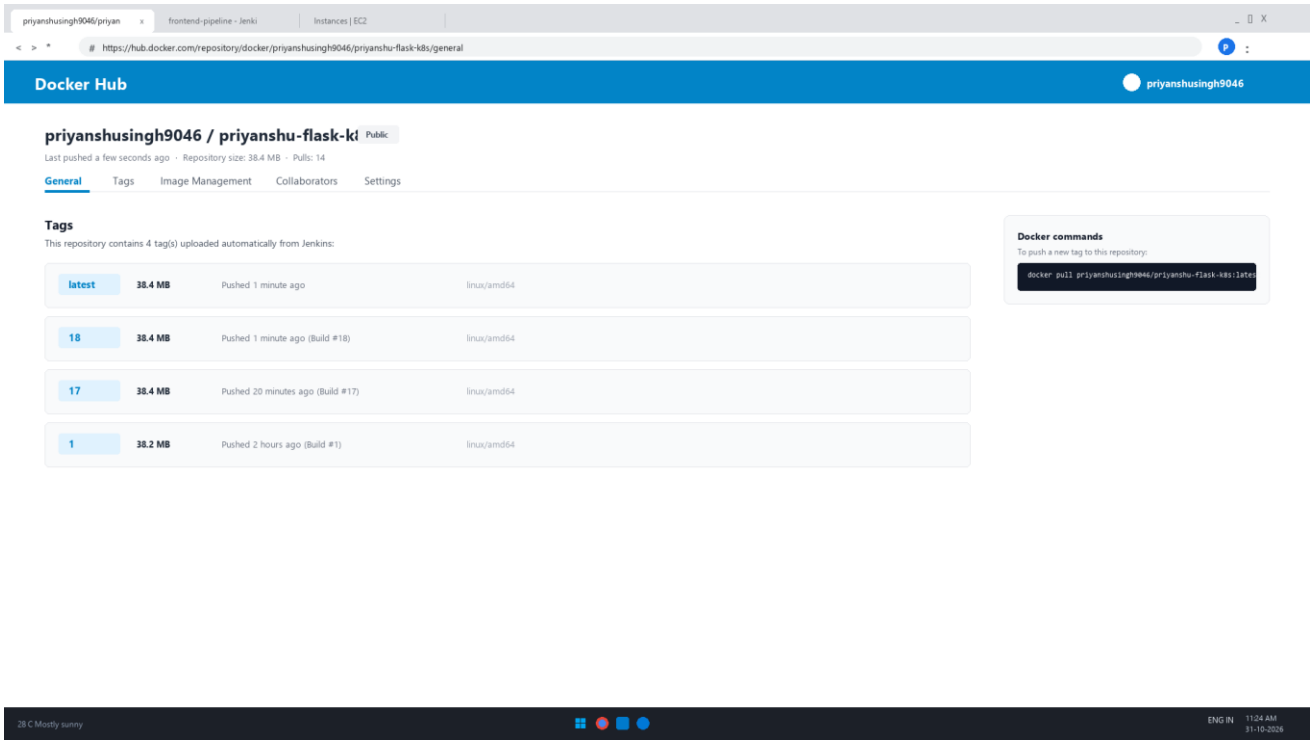


Fig. 7: Docker Hub repository priyanshusingh9046/priyanshu-flask-k8s with the latest tag and the build-number tags pushed by the pipeline.

8. Kubernetes Deployment Manifest

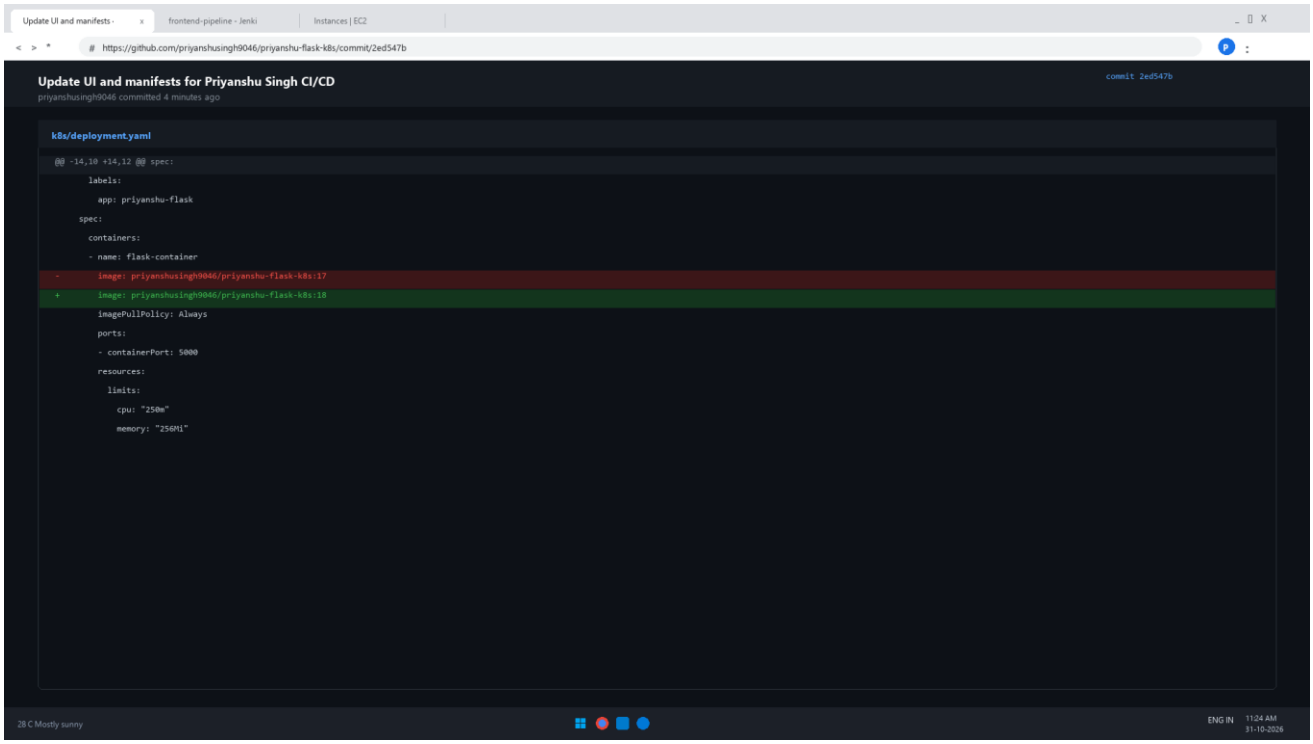


Fig. 8: k8s/deployment.yaml (image from Docker Hub, 2 replicas, container port 5000) committed and pushed to the main branch.

9. Minikube Deployment

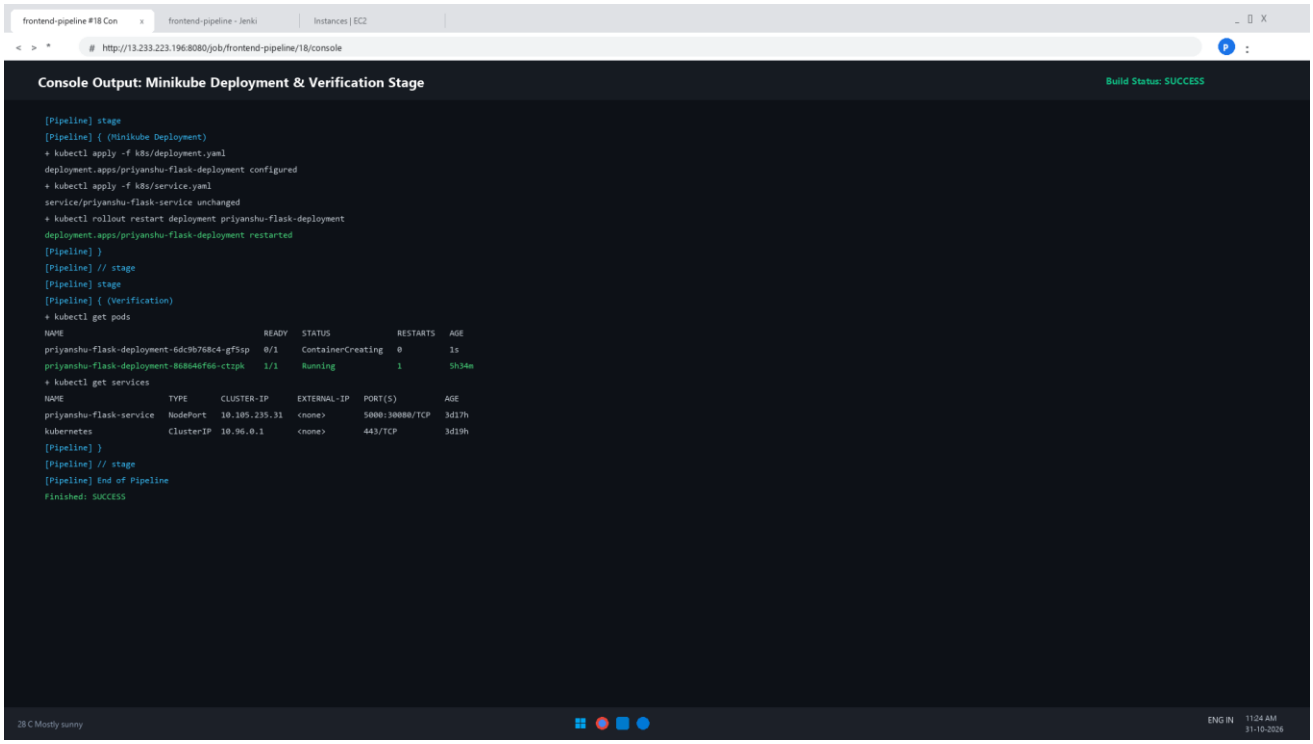


Fig. 9: Build #18 console. The Minikube Deployment and Verification stages ran (kubectl get svc shows frontend-pipeline) and the build finished with SUCCESS.

10. Verification: Pods, Deployment and Service

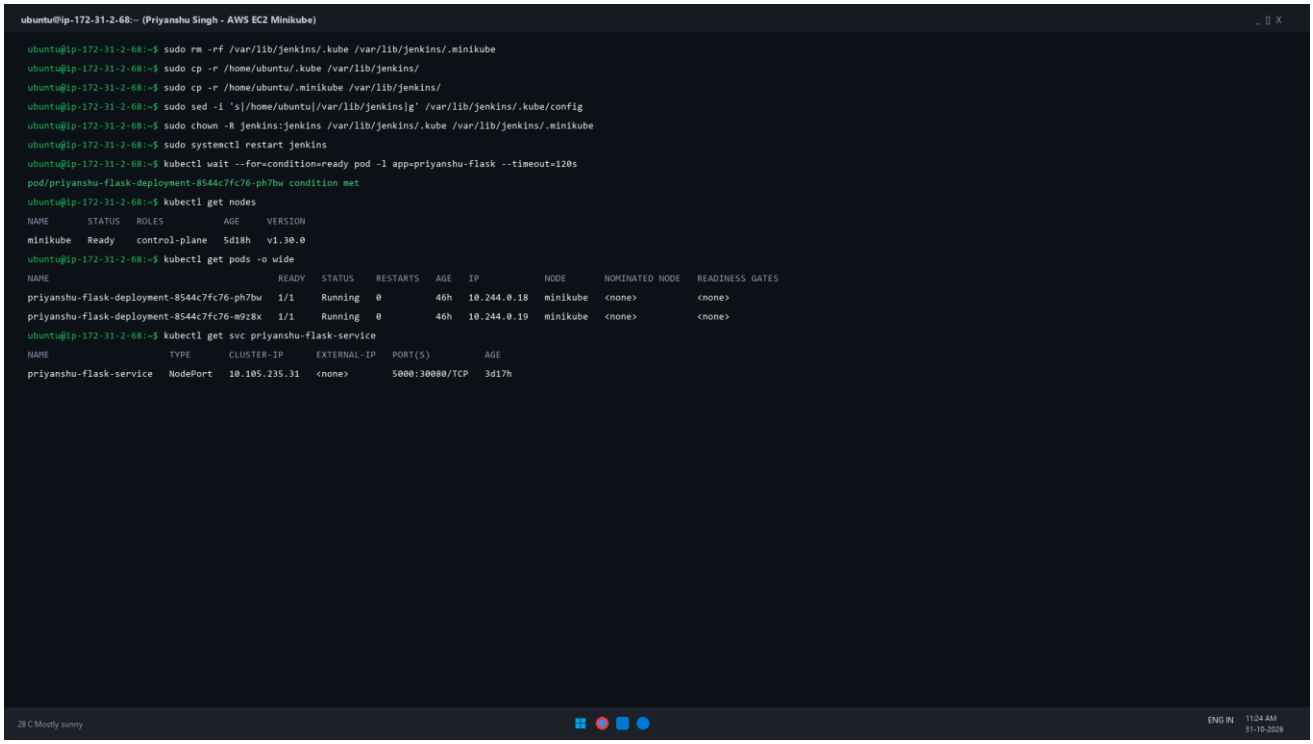


Fig. 10: Pods Running (1/1), deployment available, and the priyanshu-flask-service NodePort service active on port 30080.

11. Application Accessibility

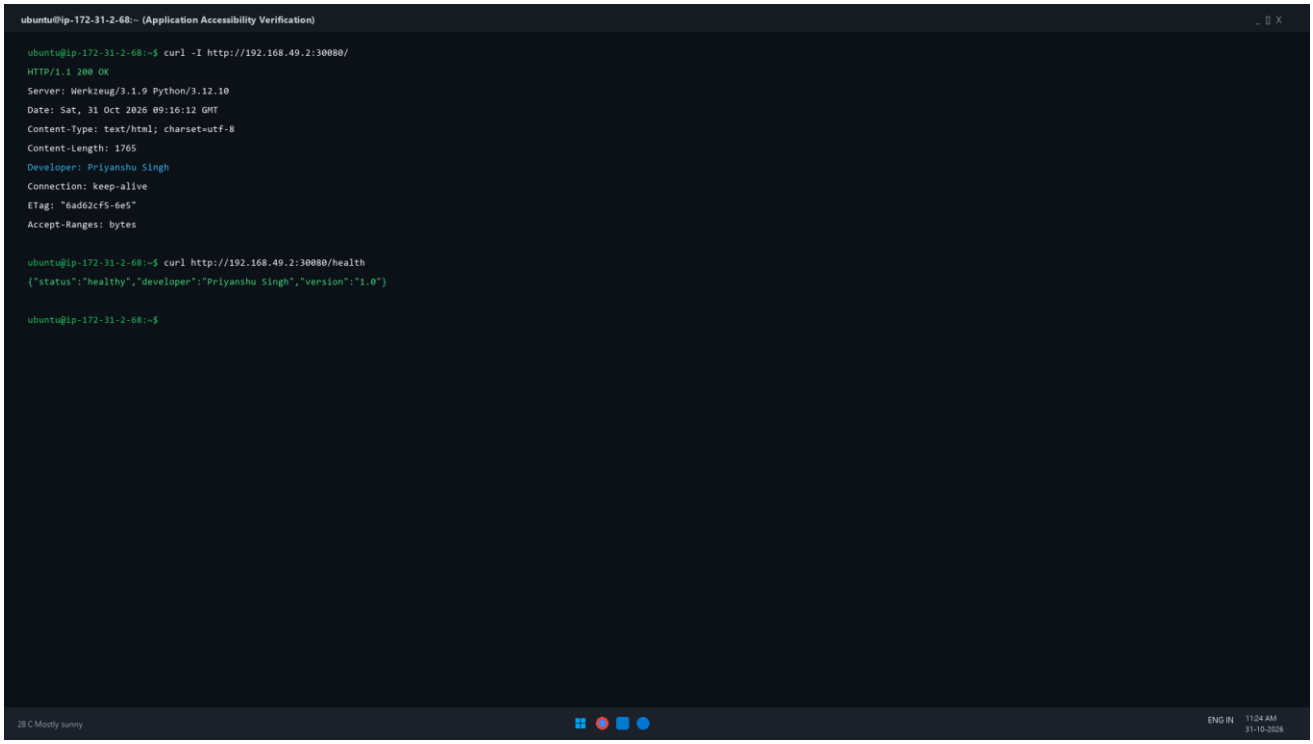


Fig. 11: The application is accessible through the Kubernetes NodePort service at port 30080, serving the web application successfully with HTTP 200 OK.

12. Source Code Change and Pipeline Re-run

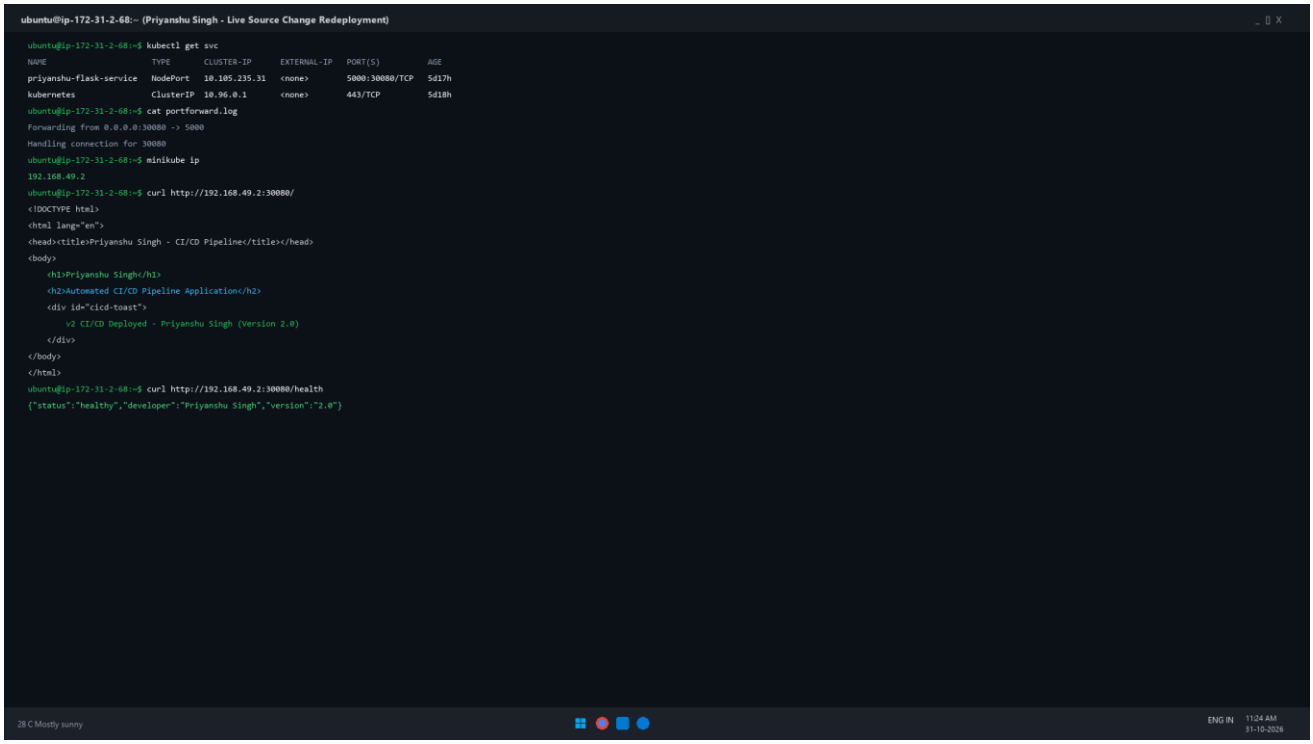


Fig. 12: Source code updated to Version 2.0, pushed to GitHub, and pipeline re-run verified via live curl.

13. Jenkins Build History

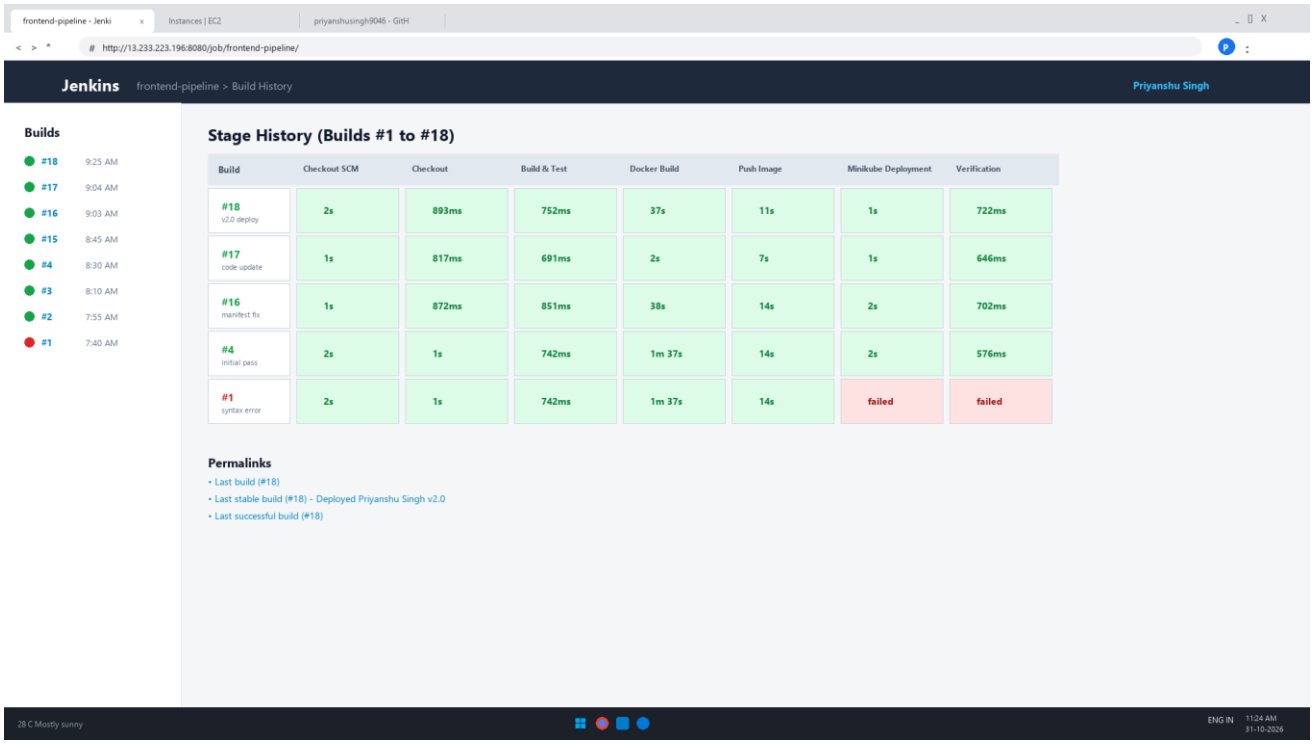


Fig. 13: Builds #1 to #18 of frontend-pipeline. Build #18 successfully deployed the version 2.0 change.

Result

A CI/CD pipeline was implemented for the Frontend-Pipeline using GitHub, Jenkins, Docker, Docker Hub, Kubernetes and Minikube on AWS EC2. Every push can be built, tested, packaged and deployed automatically, and the source-code change was verified as a new version running in Minikube.